

RADHIKA DOTIHAL

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PROFESSIONAL SUMMARY

Software engineer with 5+ years building backend systems for fintech and high-volume data processing. I focus on performance, reliability, and production debugging, with recent work reducing processing time by 98.8% and speeding up a compute-heavy workflow by 6x. Strong in C++, Java, distributed systems, low-latency architecture, and database design.

PROFESSIONAL EXPERIENCE

Software Engineer - Independent Projects

Sept 2025 - Present

Matrix Multiplication Optimizer [C++, SIMD AVX2, OpenMP]

- Optimized dense matrix multiplication in C++ by implementing cache blocking, loop unrolling, SIMD AVX2 vectorization and multi-threading, achieving 6x speedup over the naïve implementation from 0.2 GFLOPS to 1.2 GFLOPS and improving cache utilization and CPU parallelism.

AI Resume Analyzer [Python, Streamlit, Sentence Transformers]

- Developed an AI-powered resume analyzer using all-MiniLM-L6-v2 model, pdfplumber, Streamlit and Sentence Transformers which use embedding semantic similarity scoring to compute ATS match percentage between resume and job description.

URL Shortener - System Design [Python, Flask, MySQL, Redis, Docker]

- Architected and developed a URL shortening platform using Python, Flask, Redis and MySQL implementing base62 encoding for compact URL generation, low-latency caching and scalable data persistence.

Software Engineer –SS&C Technologies, Inc. Advent [C++, Python, MySQL, Linux, Docker, Jenkins, Scrum]

July 2022 – Aug 2025

- Accelerated report generation via multi-threaded code and concurrency in C++ reducing processing time by 98.8%, saving clients' valuable time and providing them with faster access to the data they need.
- Built Python automation script to validate environment configurations across DEV and PROD environment eliminating manual setup errors and reducing onboarding time for new engineers.
- Developed Python tool to detect and log configuration differences across development runs, reducing debugging time for environment inconsistencies.

Volunteer – Global Health Impact Project [Python, Flask, React.js, SQLite]

May 2021 – Aug 2021

- Implemented backend for website 'GHI Forecasting Tool' used to measure pharmaceutical product's impact on global health to increase access for medicines.

Software Engineer –LTI, India [Hyperion Infra, Oracle EPM, Java, MySQL, HTML, CSS, BootStrap, JavaScript] Aug 2018 – Nov, 2020

- Orchestrated ETL migration across UAT, DEV, and QA using Informatica PowerCenter, streamlining data workflows.
- Led OBIEE repository migrations with Oracle BI Admin, Enterprise Manager, and Data Integrator.
- Delivered a full-stack web application (Java,HTML/CSS/BootStrap/JavaScript,MySQL) enabling users to query bus schedules, select seats, and view past bookings.

ACADEMIC PROJECTS

Building & Performance Testing of a Highly Available Website using GCP Serverless Computing [JavaScript, Python, GCP]

- Deployed a website to track International Space Station and display location on Google Maps using JavaScript by utilizing features such as google cloud function, google cloud SQL and CRON job scheduler.
- Analyzed website performance by tracking the number of requests, latency & container count using Apache Benchmark tool.

Containers & Mini Docker [Python, Docker, Sysbench, QEMU, GCP]

- Implemented a Docker like tool called Mini Docker in Python to create container instances with namespace & control groups.
- Evaluated the performance of Docker container and QEMU using sysbench tool by running CPU and File I/O test modes.

Smart Home Automation using Alexa [Python, flask, AWS Lambda, XML, JSON]

- Instituted functionalities such as turning devices ON/OFF, adjusting brightness, setting schedules, creating and controlling device zones and providing security.
- Trained Alexa with intents and utterances in JSON using AWS Lambda function, maintained and tracked the database in xml.

TECHNICAL SKILLS

Programming Languages: C++, Java, Python

Web Technologies: HTML5, CSS3, JavaScript, jQuery, JSON, XML, Document Object Model (DOM)

Databases: MySQL, MongoDB, Cassandra, DynamoDB

CI/CD, Cloud & Containers: GCP, Docker, AWS (Lambda Function), Kubernetes, Jenkins, Terraform, Prometheus, Grafana

APIs: REST API, SOAP

Framework: Bootstrap, Express.js, [React.js](https://reactjs.org/), JUnit, Mocha-Chai, Flask, LangChain, CrewAI

AI/ML & LLM Systems: OpenAI ChatGPT, Claude, LLM Orchestration, RAG, Cursor AI

Version Control: GitHub

Tools: VS Code, Eclipse, Jupyter Notebook, Apache Kafka

Methodologies: Test-Driven Development, Scrum, Agile

EDUCATION

Master of Science, Computer Science State @ University of New York at Binghamton, New York

Feb 2021 – May 2022

Bachelor of Engineering, Information Technology @ University of Pune, India

Aug 2014 – May 2018

PUBLICATIONS & ACHIEVEMENTS

- Presented and published research paper titled “[*Smart Homes Using Alexa and Power Line Communication in IoT | SpringerLink*](#)” at an annual ICCCNT conference in Springer. **April,2018**
- Completed online certification in AI Agents, ChatBot Development, LLM fundamentals & AI Ethics. **May 2026**
- Pursued self-paced learning in Terraform, Grafana, Apache Kafka 101 and Apache Kafka Connect 101. **May 2026**